

INTERNATIONAL STANDARD

Fibre optic communication subsystem test procedures – Part 2-3: Digital systems – Jitter and wander measurements



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CONTENTS

FOREWORD.....	5
1 Scope.....	7
1.1 Types of jitter measurements	7
1.2 Types of wander measurements	7
2 Normative references	7
3 Terms and definitions	7
4 General considerations.....	11
4.1 Jitter generation	11
4.1.1 Timing jitter	11
4.1.2 Alignment jitter	11
4.1.3 Other effects.....	12
4.2 Effects of jitter on signal quality.....	12
4.3 Jitter tolerance	12
4.4 Waiting time jitter	13
4.5 Wander	14
5 Jitter test procedures.....	14
5.1 General considerations.....	14
5.1.1 Analogue method	14
5.1.2 Digital method	14
5.2 Common test equipment.....	15
5.3 Safety	16
5.4 Fibre optic connections	17
5.5 Test sample	17
6 Jitter tolerance measurement procedure.....	17
6.1 Purpose	17
6.2 Apparatus.....	17
6.3 BER penalty technique	17
6.3.1 Equipment connection	17
6.3.2 Equipment settings	18
6.3.3 Measurement procedure.....	18
6.4 Onset of errors technique	18
6.4.1 Equipment connection	18
6.4.2 Equipment settings	19
6.4.3 Measurement procedure.....	19
6.5 Jitter tolerance stressed eye receiver test	20
6.5.1 Purpose.....	20
6.5.2 Apparatus.....	20
6.5.3 Sinusoidal jitter template technique	20
7 Measurement of jitter transfer function	21
7.1 General.....	21
7.2 Apparatus.....	21
7.3 Basic technique.....	22
7.3.1 Equipment connection	22
7.3.2 Equipment settings	22
7.3.3 Measurement procedure.....	22
7.4 Analogue phase detector technique.....	23

7.4.1	Equipment connections.....	23
7.4.2	Equipment settings	23
7.4.3	Measurement procedure	24
7.4.4	Measurement calculations	24
8	Measurement of output jitter	24
8.1	General	24
8.2	Equipment connection	24
8.2.1	Equipment settings	24
8.2.2	Measurement procedure	24
8.2.3	Controlled data	25
9	Measurement of systematic jitter	25
9.1	Apparatus.....	25
9.2	Basic technique	25
9.2.1	Equipment connection	25
9.2.2	Equipment settings	26
9.2.3	Measurement procedure	26
10	BERT scan technique	27
10.1	Apparatus.....	29
10.2	Basic technique	29
10.2.1	Equipment connection	29
10.2.2	Equipment settings	29
10.2.3	Measurement process	29
11	Jitter separation technique	30
11.1	Apparatus.....	31
11.2	Equipment connections	31
11.3	Equipment settings.....	31
11.4	Measurement procedure.....	32
11.4.1	Sampling oscilloscope:	32
11.4.2	Real-time oscilloscope.....	32
12	Measurement of wander	33
12.1	Apparatus.....	33
12.2	Basic technique	33
12.2.1	Equipment connection	33
12.2.2	Equipment settings	34
12.2.3	Measurement procedure	35
13	Measurement of wander TDEV tolerance.....	35
13.1	Intent.....	35
13.2	Apparatus.....	35
13.3	Basic technique	35
13.4	Equipment connection	35
13.4.1	Wander TDEV tolerance measurement for the test signal of EUT	35
13.4.2	Wander TDEV tolerance measurement for timing reference signal of EUT	36
13.5	Equipment settings.....	36
13.6	Measurement procedure.....	37
14	Measurement of wander TDEV transfer	37
14.1	Apparatus.....	37
14.2	Equipment connection	37

14.2.1 Wander TDEV transfer measurement for the test signal of EUT	37
14.2.2 Wander TDEV transfer measurement for timing reference signal of EUT	37
14.3 Equipment settings.....	38
14.4 Measurement procedure.....	38
15 Test results	38
15.1 Mandatory information.....	38
15.2 Available information.....	39
Bibliography.....	40
Figure 1 – Jitter generation	11
Figure 2 – Example of jitter tolerance.....	13
Figure 3 – Jitter and wander generator	15
Figure 4 – Jitter and wander measurement	16
Figure 5 – Jitter stress generator	16
Figure 6 – Jitter tolerance measurement configuration: bit error ratio (BER) penalty technique.....	18
Figure 7 – Jitter tolerance measurement configuration: Onset of errors technique	19
Figure 8 – Equipment configuration for stressed eye tolerance test.....	20
Figure 9 – Measurement of jitter transfer function: basic technique.....	22
Figure 10 – Measurement of Jitter transfer: analogue phase detector technique	23
Figure 11 – Output jitter measurement.....	25
Figure 12 – Systematic jitter measurement configuration: basic technique	26
Figure 13 – Measurement of the pattern-dependent phase sequence xi	27
Figure 14 – BERT scan bathtub curves (solid line for low jitter, dashed line for high jitter).....	28
Figure 15 – Equipment setup for the BERT scan.....	29
Figure 16 – Dual Dirac jitter model.....	31
Figure 17 – Equipment setup for jitter separation measurement.....	31
Figure 18 – Measurement of time interval error.....	32
Figure 19 – Synchronized wander measurement configuration.....	34
Figure 20 – Non-synchronized wander measurement configuration.....	34
Figure 21 – Wander TDEV tolerance measurement configuration for the test signal of EUT	36
Figure 22 – Wander TDEV tolerance measurement configuration for the timing signal of EUT	36
Figure 23 – Wander TDEV transfer measurement configuration for the test signal of EUT	37
Figure 24 – Wander TDEV transfer measurement configuration for the timing signal of EUT	38

INTERNATIONAL ELECTROTECHNICAL COMMISSION

FIBRE OPTIC COMMUNICATION SUBSYSTEM TEST PROCEDURES –

Part 2-3: Digital systems – Jitter and wander measurements

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The text of this standard is based on the following documents:

FDIS	Report on voting
86C/885/FDIS	86C/905/RVD

Full information on the voting for the approval of this standard can be found in the report on voting indicated in the above table.

A list of all parts of the IEC 61280-2 series, published under the general title *Fibre optic communication subsystem test procedures – Digital systems*, can be found on the IEC website.

The committee has decided that the contents of this publication will remain unchanged until the maintenance result date indicated on the IEC web site under "<http://webstore.iec.ch>" in the data related to the specific publication. At this date, the publication will be

- reconfirmed;
- withdrawn;
- replaced by a revised edition, or
- amended.

A bilingual version may be published at a later date.

FIBRE OPTIC COMMUNICATION SUBSYSTEM TEST PROCEDURES –

Part 2-3: Digital systems – Jitter and wander measurements

1 Scope

This part of IEC 61280 specifies methods for the measurement of the jitter and wander parameters associated with the transmission and handling of digital signals.

1.1 Types of jitter measurements

This standard covers the measurement of the following types of jitter parameters:

- a) jitter tolerance
 - 1) sinusoidal method
 - 2) stressed eye method
- b) jitter transfer function
- c) output jitter
- d) systematic jitter
- e) jitter separation

1.2 Types of wander measurements

This standard covers the measurement of the following types of wander parameters:

- a) non-synchronized wander
- b) TDEV tolerance
- c) TDEV transfer
- d) synchronized wander

2 Normative references

The following referenced documents are indispensable for the application of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

IEC 60825-1, *Safety of laser products – Part 1: Equipment classification and requirements*

ITU-T Recommendation G.813, *Timing characteristics of SDH equipment slave clocks (SEC)*

3 Terms and definitions

For the purposes of this document, the following terms and definitions apply.

NOTE See also IEC 61931.

3.1

jitter

the short-term, non-cumulative, variation in time of the significant instances of a digital signal from their ideal position in time. Short-term variations in this context are jitter components with a repetition frequency equal to or exceeding 10 Hz

3.2

jitter amplitude

the deviation of the significant instance of a digital signal from its ideal position in time

NOTE For the purposes of this standard the jitter amplitude is expressed in terms of the unit interval (UI). It is recognized that jitter amplitude may also be expressed in units of time.

3.3

unit interval (UI)

the shortest interval between two equivalent instances in ideal positions in time. In practice this is equivalent to the ideal timing period of the digital signal

3.4

jitter frequency

the rate of variation in time of the significant instances of a digital signal relative to their ideal position in time. Jitter frequency is expressed in Hertz (Hz)

3.5

jitter bandwidth

the jitter frequency at which the jitter amplitude has decreased by 3dB relative to its maximum value

3.6

alignment jitter

jitter created when the timing of a data signal is recovered from the signal itself

3.7

timing jitter

jitter present on a timing source

3.8

systematic jitter

jitter components which are not random and have a predictable rate of occurrence. Systematic jitter in a digital signal results from regularly recurring features in the digital signal, such as frame alignment data, and justification control data. This is sometimes referred to as deterministic jitter and is composed of periodic uncorrelated jitter and data dependent jitter

3.9

periodic uncorrelated jitter

a form of systematic jitter that occurs at a regular rate, but is uncorrelated to the data when the data pattern repeats. Periodic uncorrelated jitter will be the same independent of which edge in a pattern is observed over time. Sources of periodic uncorrelated jitter include switching power supplies phase modulating reference clocks or any form of periodic phase modulation of clocks that control data rates

3.10

inter-symbol interference jitter

caused by bandwidth limitations in transmission channels. If the channel bandwidth is low, signal transitions may not reach full amplitude before transitioning to a different logic state. Starting at a level closer to the midpoint between logic states, the time at which the signal edge then crosses a specific amplitude threshold can be early compared to consecutive identical digits which have reached full amplitude and then switch to the other logic state

3.11**duty cycle distortion**

occurs when the duration of a logic 1 (0-1-0) is different from the duration of a logic 0 (1-0-1). For example, if the logic 1 has a longer duration, rising edges will occur early relative to falling edges, compared to their ideal locations in time

3.12**data dependent jitter**

represents jitter that is correlated to specific bits in a repeating data pattern. That is, when a data pattern repeats, the jitter on any given signal edge will manifest itself in the same way for any repetition of the pattern. It is due to either duty cycle distortion and/or inter-symbol interference

3.13**waiting time jitter**

applies to plesiochronous multiplexing and is defined as the jitter caused by the varying delay between the demand for justification and its execution

3.14**jitter tolerance**

maximum jitter amplitude that a digital receiver can accept for a given penalty or alternatively without the addition of a given number of errors to the digital signal. The maximum jitter amplitude tolerated is generally dependent on the frequency of the jitter

3.15**jitter generation**

process of adding jitter impairment to a data signal

3.16**input jitter**

magnitude of the jitter occurring at a hierarchical interface or the input port of equipment or a device

3.17**output jitter**

magnitude of the jitter occurring at a hierarchical interface or the output port of equipment or a device

3.18**jitter transfer**

amount of jitter transferred from the input to the output of an equipment or device. It is usually expressed as a ratio (in dB) of the output jitter to the input jitter

3.19**total jitter**

the summation (or convolution) of deterministic and random jitter. Total jitter is expressed as a peak value

3.20**jitter bathtub curve**

display of bit-error-ratio as a function of the time location of the BERT error detector sampling point. The resulting curve is then a display of the probability that a data edge will be misplaced at or beyond a specific location (closer to the centre of a bit) within a unit interval

3.21**wander**

long-term, non-cumulative, variation in time of the significant instances of a digital signal from their ideal position in time. Long-term variations in this context are jitter components with a repetition frequency less than 10 Hz

NOTE For the purpose of this document, the wander amplitude is expressed in units of time (s). It is recognised that wander amplitude may also be expressed in terms of unit interval (UI).

3.22

time interval error

TIE

difference between the measure of a time interval as provided by a clock and the measure of that same time interval as provided by a reference clock. Mathematically, the time interval error function TIE ($t; \tau$) can be expressed as:

$$TIE(t; \tau) = [T(t + \tau) - T(t)] - [T_{ref}(t + \tau) - T_{ref}(t)] = x(t + \tau) - x(t) \quad (1)$$

where τ is the time interval, usually called observation interval

3.23

maximum time interval error

MTIE

maximum peak-to-peak delay variation of a given timing signal with respect to an ideal timing signal within an observation time ($\tau = n\tau_0$) for all observation times of that length within the measurement period (T). It is estimated using the following formula:

$$MTIE(n\tau_0) \equiv \max_{1 \leq k \leq N-n} \left[\max_{k \leq i \leq k+n} x_i - \min_{k \leq i \leq k+n} x_i \right] \quad n = 1, 2, \dots, N-1 \quad (2)$$

3.24

time deviation

TDEV or σ_x

measure of the expected time variation of a signal as a function of integration time. TDEV can also provide information about the spectral content of the phase (or time) noise of a signal. TDEV is in units of time. Based on the sequence of time error samples, TDEV is estimated using the following calculation:

$$TDEV(n\tau_0) \equiv \sqrt{\frac{1}{6n^2(N-3n+1)} \sum_{j=1}^{N-3n+1} \left[\sum_{i=j}^{n+j-1} (x_{i+2n} - 2x_{i+n} + x_i) \right]^2} \quad n = 1, 2, \dots, \text{int} \left(\frac{N}{3} \right) \quad (3)$$

where

x_i denotes time error samples;

N denotes the total number of samples;

τ_0 denotes the time error-sampling interval;

τ denotes the integration time, the independent variable of TDEV;

n denotes the number of sampling intervals within the integration time t .

3.25

bit error ratio

BER

number of bits received in error as a ratio of the total number of bits received

3.26

errored second

time of 1 s duration that contains one or more digital errors in a data stream

4 General considerations

4.1 Jitter generation

Jitter in a digital signal is generated by three basic processes which are briefly described below. The mathematical analysis of the jitter processes is complex and not within the scope of this standard. A comprehensive analysis and early mathematical treatise of the jitter processes is provided by [1]¹.

4.1.1 Timing jitter

Jitter impairment of the original data timing clock. Even the most stable timing sources contain a certain amount of jitter, or unintended phase modulation or phase noise. In primary timing generators this impairment is exceedingly small, but is increased when such timing signals are distributed in a system. The effect of noise on the timing signal in a digital system is demonstrated in an exaggerated degree in Figure 1.

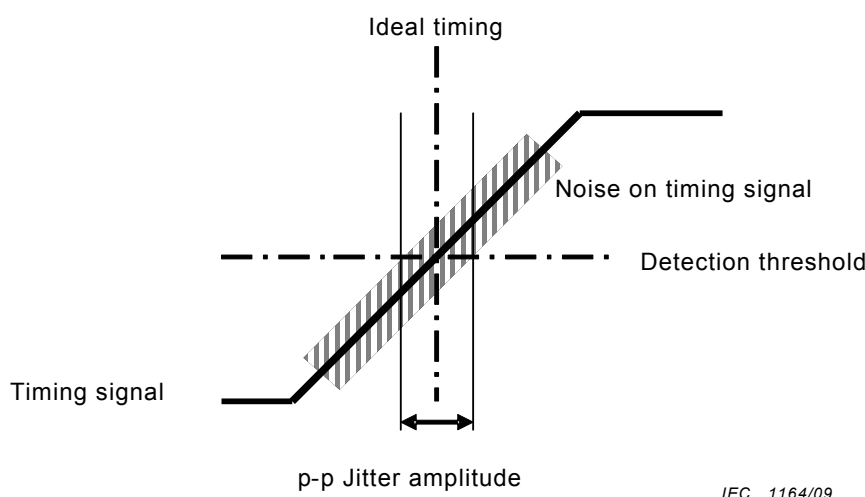


Figure 1 – Jitter generation

4.1.2 Alignment jitter

When a digital pattern is presented to a timing recovery circuit, the continuous variation of the digital pattern results in the creation of jitter in the recovered clock signal relative to the incoming data (alignment jitter). This effect, first analyzed and described in detail by [2] is the major cause of jitter generation. It means that the jitter components of the recovered clock signal are added to the data when they are retimed. The jitter bandwidth created by this process is the same as the analogue bandwidth of the clock recovery circuit used.

When the process is repeated at a similar equipment, the resultant clock signal shows increased jitter due to the addition of timing and alignment jitter. Thus, jitter is added to the data signal, and amplified at the next timing recovery operation. A repetition of this process, such as occurs in transmission links with many repeaters, or chains of add-drop multiplexers, can build up substantial jitter amplitudes; but as long as the bandwidth of the timing recovery process is the same or greater than the jitter bandwidth of the signal, the jitter will always be accommodated. An analysis of the accumulation of jitter in successive timing recovery operations was first published by [3]. The jitter buildup can be represented by an equation of the form:

¹ Figures in square brackets refer to the Bibliography.

$$\theta_n = \sum_{k=1}^n \theta_0(j\omega) \left(\frac{1}{1 + j\omega/B} \right)^k \quad (4)$$

The above equation yields a jitter power density spectrum which can be expressed in the form:

$$\phi_n \approx n^2 \left[\frac{\sin \frac{n}{2B} \omega}{\frac{n}{2B} \omega} \right]^2 \phi_0 \text{ for } \omega \ll B \quad (5)$$

where

- θ_n denotes the jitter amplitude after n timing recovery processes
- θ_0 denotes the jitter amplitude introduced at each individual timing recovery process
- n denotes the number of tandem timing recovery processes
- ω denotes the angular frequency ($2\pi f$) of the jitter component
- B denotes the half angular bandwidth of the timing recovery circuit
- ϕ_n denotes the jitter power density after n timing recovery processes
- ϕ_0 denotes the jitter power density introduced at each individual timing recovery process

It should be noted that for low values of frequency the power density and hence amplitude of the jitter increases linearly with the number of tandem timing recovery processes.

In point-to-point communications systems, where transmitter timing is not derived from incoming data, alignment jitter and jitter build-up is not a significant problem.

4.1.3 Other effects

In the course of the transmission of a digital signal, further impairments such as added noise and dispersion effects provide additional jitter components when timing is recovered from the signal. Such effects are more severe when analogue amplification is used rather than digital regeneration in order to increase the length of a digital link.

4.2 Effects of jitter on signal quality

Jitter has no effect on the transmission of data as long as the equipment can accommodate the jitter amplitude and rate of deviation (see 4.3). When jitter is large enough or fast enough such that the receiver decision point is made near or beyond a data edge, a mistake can be made and BER degraded. Jitter, depending on its amplitude and frequency, can also have serious effects on analogue services such as music and television which have been transmitted over digital links. The effect of jitter is to introduce unwanted frequency and phase modulation products which are audible in music and visible on television pictures.

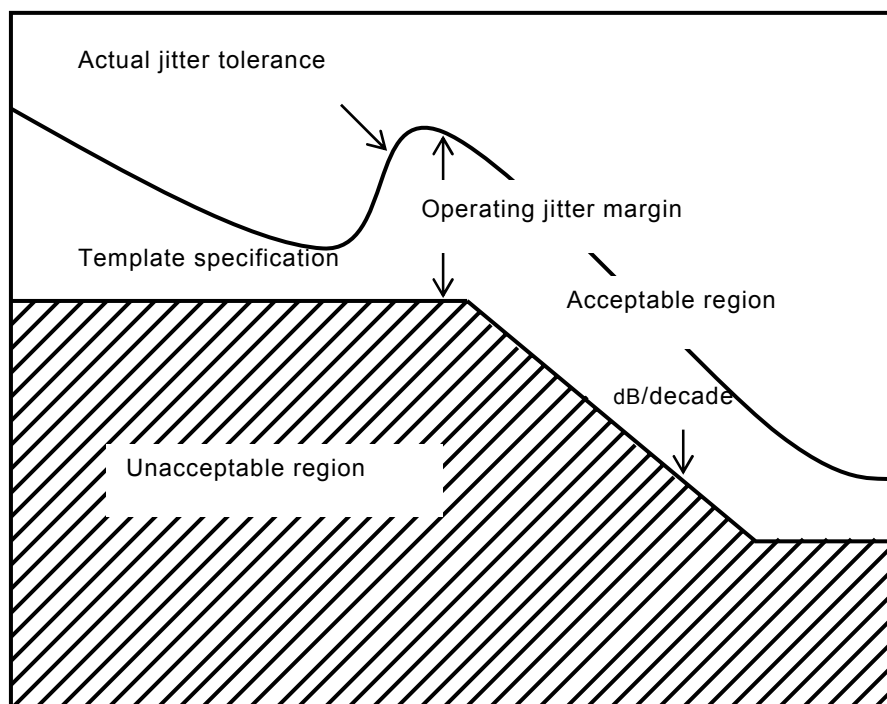
4.3 Jitter tolerance

In telecommunications systems, jitter tolerance requirements are typically specified in terms of jitter templates, which cover a specified sinusoidal amplitude/frequency region. Jitter templates represent the minimum amount of jitter the equipment shall be able to accept without producing the specified degradation of error performance. A typical relationship between actual jitter tolerance and its associated tolerance template is illustrated in Figure 2.

The jitter amplitudes that equipment actually tolerates at a given frequency are defined as all amplitudes up to, but not including, that which causes the designated degradation of error performance. The designated degradation of error performance may be expressed in terms of either bit-error-ratio (BER) penalty or the onset of errors criteria. The existence of these two

criteria arises because the input jitter tolerance of digital equipment is primarily determined by the following three factors:

- ability of the input clock recovery circuit to accurately recover the timing from a jittered data signal, including the presence of other degradations such as pulse distortion, crosstalk, noise, and other impairments;
- ability of the input circuit buffer, for example an elastic store, to accommodate the jitter amplitude;
- ability of other components to accommodate dynamically varying input data rates such as pulse justification capacity and synchronizer and de-synchronizer buffer size in an asynchronous digital multiplex.



IEC 1165/09

Figure 2 – Example of jitter tolerance

In data communications systems, jitter tolerance is often determined with signal impairments that are more complex than simple sinusoidal jitter. The general concept is to verify that the receiver is capable of achieving the desired BER when presented with the allowable signal it will encounter in a real system. Thus the jitter tolerance test signal will include impairments that are allowed for both the transmitter and the channel. For example, a real transmitter may have periodic jitter, random jitter, and duty cycle distortion. As the signal traverses the channel, it may be further degraded through a bandwidth limited channel, thus adding inter-symbol interference jitter. As the receiver shall be able to tolerate such a signal in a real system, the signal used to verify receiver tolerance shall include all of these impairments. This method of testing is sometimes referred to as “stressed eye” testing, indicating that the eye diagram of the signal presented to a receiver has been intentionally degraded or stressed.

4.4 Waiting time jitter

When asynchronous (plesiochronous) signals are multiplexed, a justification technique (also known as pulse stuffing) is used which involves the comparison of the phase of the incoming digital signal with the multiplexer’s tributary timing. When a preset difference is detected a control signal is transmitted, via the overhead in the multiplex frame structure, to the demultiplexer. In order to ensure the integrity of the control signal in the presence of errors, it is repeated 3 or 5 times. At the demultiplexer a majority decision is taken to recognize the

control signal. This delay introduces an uncertainty and varying delay in the time between the demand for justification and its execution at the demultiplexer, and expresses itself as waiting time jitter.

4.5 Wander

Wander is essentially caused by periodic variations in the delay of a transmission path and slow variations in the frequency of data clocks. The boundary between jitter and wander at a frequency of 10 Hz is somewhat artificial since most significant wander effects occur at repetition rates of hours, days, months and seasons. In general wander is caused by temperature changes which affect the delays of the transmission medium or equipment. For terrestrial transmission, wander components are normally limited to a few tens of nanoseconds.

In synchronous networks the accommodation of wander is an essential feature since a single bit slip will initiate a resynchronization process with consequent loss of data. If this occurs at a high level demultiplexer, all lower levels will also resynchronize which will exacerbate the loss of data. Wander is normally accommodated by increasing the size of the elastic store used to accommodate jitter.

A special case is the wander effect that occurs in communications via geostationary satellites. While the lateral position of such satellites is relatively stable, their mean height of some 35 000 km above the earth's surface has a diurnal variation which may be up to approximately $\pm 1\,000$ km. This results in delay variations per hop of approximately 26 ms. At a data rate of 150 Mbit/s this represents a wander rate of some 45 bits/s and requires an elastic store with a capacity of at least 3,9 Mbits to accommodate it in a synchronous network.

5 Jitter test procedures

5.1 General considerations

In order to measure jitter, analogue and digital methods may be used. Both methods rely on the phase comparison between a recovered timing signal representing the signal to be measured, or in some cases the signal itself, and a stable clock signal whose frequency represents the ideal signal. In telecommunications applications, this ideal signal is derived from the long-term average frequency of the derived timing signal. In order to obtain meaningful results, the jitter bandwidth of the derived clock shall be significantly less than 10 Hz. In data communications applications, the reference clock is also typically derived from the signal, but the jitter bandwidth of the derived signal will often be similar to the clock recovery bandwidth of the receiver for which a transmitter under test will be paired with. In system use, jitter that is within the loop bandwidth of the receiver is accommodated by the receiver and of lower concern. In test, jitter that is common to both the reference signal and the signal to be measured may intentionally not be observed. This facilitates the ability to observe jitter outside the jitter bandwidth.

5.1.1 Analogue method

The analogue method uses the analogue output from a phase comparison between the recovered clock and the stable clock which provides a pulse width modulated signal presentation of the jitter components in terms of amplitude and frequency. This is subsequently converted into an analogue output which is used to process the results. Like all analogue measuring techniques, this method requires careful calibration and is highly dependant on the performance characteristics, including stability, of the phase comparator.

5.1.2 Digital method

5.1.2.1 Derived clock versus narrow jitter bandwidth clock

The digital method uses a very high speed sampling clock to measure the time difference between the significant instances of the recovered and the derived stable clock signals. The

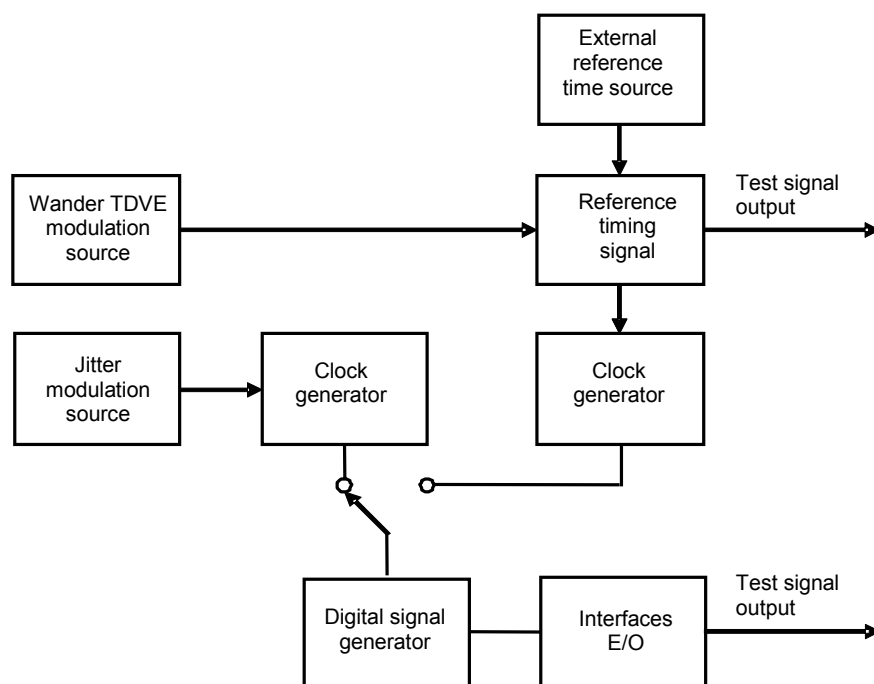
results are then obtained by digital processing of the measured time intervals. This method is capable of providing essentially accurate results and is naturally suited for measuring equipment using digital techniques. The main difficulty with this method is to provide a timing signal with a sufficiently high rate to obtain sufficient resolution.

5.1.2.2 Data edge versus reference or extracted clock

With a reference clock representing the ideal position of data edges, a population of edge locations relative to ideal is collected. This population is then post-processed to determine the jitter performance. Deterministic and random jitter contributions can be isolated. Further processing allows the estimate of the aggregate jitter (total jitter) to extremely low probabilities without the long measurement time required to make a direct measurement to low probabilities. This technique is capable of making very accurate measurements and estimations. The main difficulty with this technique is in misinterpreting low probability deterministic jitter as random jitter, which in turn leads to pessimistic estimates of total jitter. Test system bandwidth limitations and noise can also corrupt jitter results, as they can act as sources of jitter.

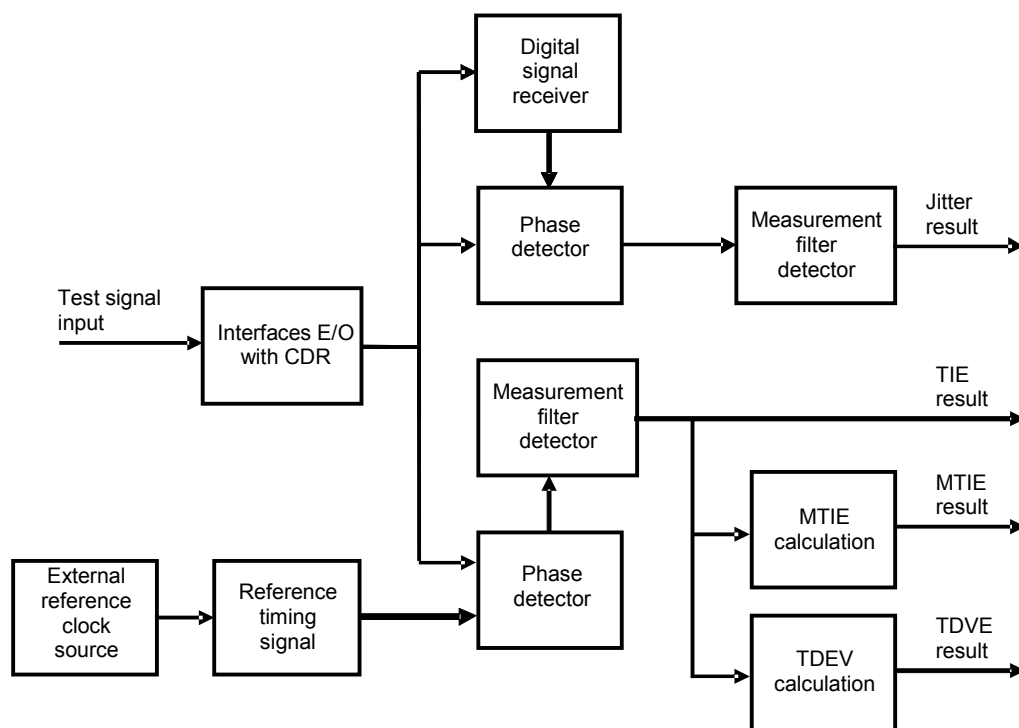
5.2 Common test equipment

Figures 3, 4 and 5 show the block diagrams of the common test equipment in general form, identifying the main functions that are used for the measurements described in this standard. The figures do not imply a specific method of implementation.



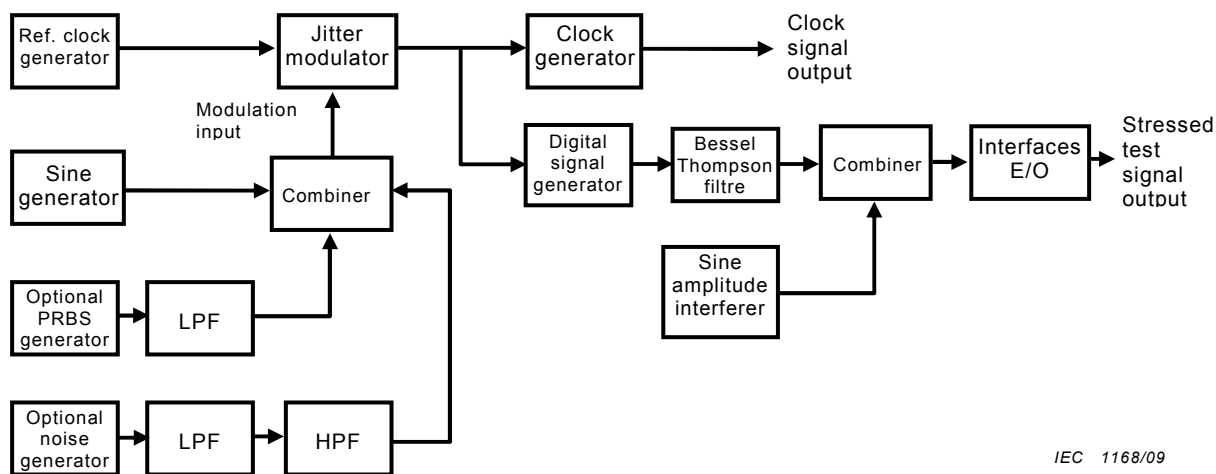
IEC 1166/09

Figure 3 – Jitter and wander generator



IEC 1167/09

Figure 4 – Jitter and wander measurement



IEC 1168/09

Figure 5 – Jitter stress generator

5.3 Safety

All tests, performed on optical fibre communication systems, or that use a laser or light emitting diodes in a test set, shall be carried out with safety precautions in accordance with IEC 60825-1.

5.4 Fibre optic connections

Fibre optic connections for the purpose of measurements shall be made using appropriate fibre optic test cords.

Before connecting equipment all connectors shall be cleaned.

In cases where high power optical signals are employed the bending radius of the test cords shall be greater than 30 mm.

5.5 Test sample

The sample is the specified equipment or transmission path under test.

6 Jitter tolerance measurement procedure

6.1 Purpose

The purpose of this test procedure is to measure jitter tolerance (also known as jitter accommodation) in terms of the sinusoidal jitter amplitude that, when applied to an equipment input, causes a designated degradation of error performance. Jitter tolerance is a function of the amplitude and frequency of the applied jitter.

6.2 Apparatus

The following apparatus is required:

- jitter generator
- digital signal generator
- digital signal receiver
- attenuator

The following apparatus is optional:

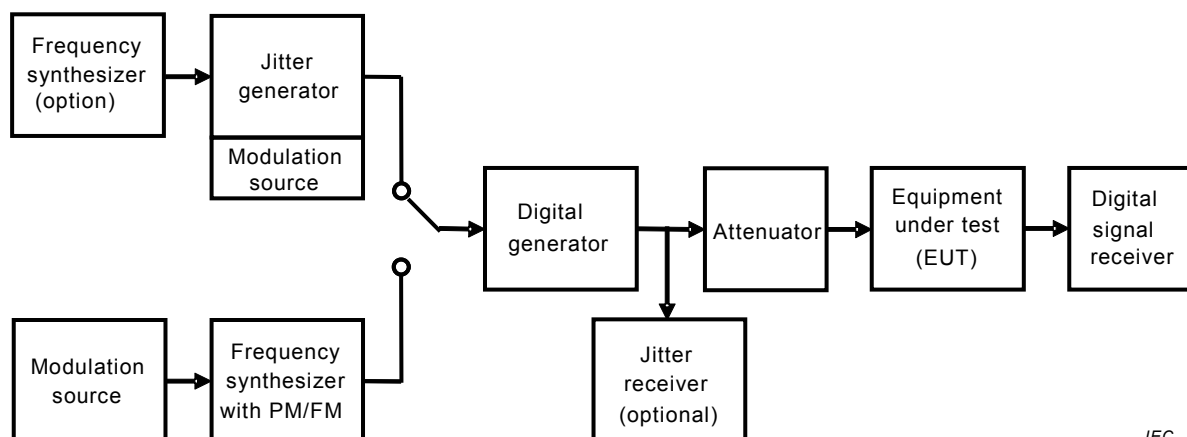
- frequency synthesizer
- jitter receiver

6.3 BER penalty technique

The bit error ratio (BER) penalty criterion for jitter tolerance measurements is defined as the amplitude of jitter, at a given jitter frequency, that duplicates the BER degradation caused by a specified signal-to-noise ratio (SNR) reduction.

6.3.1 Equipment connection

Figure 6 illustrates the test configuration for the BER penalty technique. The optional frequency synthesizer is used to provide a more accurate determination of frequencies utilized in the measurement procedure. This may be particularly important for the repeatability of measurements for some types of equipment i.e. asynchronous digital multiplexers. The optional jitter receiver is used to verify the amplitude of generated jitter.



IEC 1169/09

**Figure 6 – Jitter tolerance measurement configuration:
bit error ratio (BER) penalty technique**

6.3.2 Equipment settings

This technique is separated into two parts. Part one determines two BER versus SNR reference points for the equipment under test. With zero jitter applied, the signal is attenuated, until a convenient initial BER is obtained. Then signal attenuation is decreased until the SNR at the decision circuit is increased by the specified amount of dB (i.e. 1 dB). Part two uses the BER versus SNR reference points; at a given frequency, jitter is added to the test signal until the BER returns to its initially selected value. Since a known decision circuit eye width margin was established by the two BER versus SNR points, the added equivalent jitter is a true and repeatable measure of the decision circuit jitter tolerance performance. Part two of the technique is repeated for a sufficient number of frequencies such that the measurement accurately represents the continuous sinusoidal input jitter tolerance of the EUT over the applicable frequency range. The test equipment shall be able to produce a controlled jittered signal, a controlled SNR on the data stream, and measure the resulting BER from the EUT.

6.3.3 Measurement procedure

- Connect the equipment as shown in Figure 6. Verify proper continuity and error-free operation.
- With no applied jitter, increase the attenuation of the signal until at least 100 bit errors per second are observed. (i.e. Error Rate = $1E-10$ for G.8251).
- Record the corresponding BER and its associated SNR.
- Increase the SNR by the specified amount (i.e. 1 dB).
- Set the input jitter frequency as desired.
- Adjust the jitter amplitude until the BER returns to the value recorded in step c).
- Record the amplitude and frequency of the applied input jitter, and repeat steps e) to g) for a sufficient number of frequencies to characterize the jitter tolerance curve.

6.4 Onset of errors technique

The onset of errors criterion for jitter tolerance measurements is defined as the largest amplitude of jitter at a specified frequency that causes a cumulative total of no more than 2 errored seconds, where these errored seconds have been summed over successive 30 s measurement intervals of increasing jitter amplitude.

6.4.1 Equipment connection

Figure 7 illustrates the test configuration for the onset of errors technique. The optional frequency synthesizer is used to provide a more accurate determination of frequencies

utilized in the measurement procedure. The optional jitter measuring circuit is used to verify the amplitude of generated jitter.

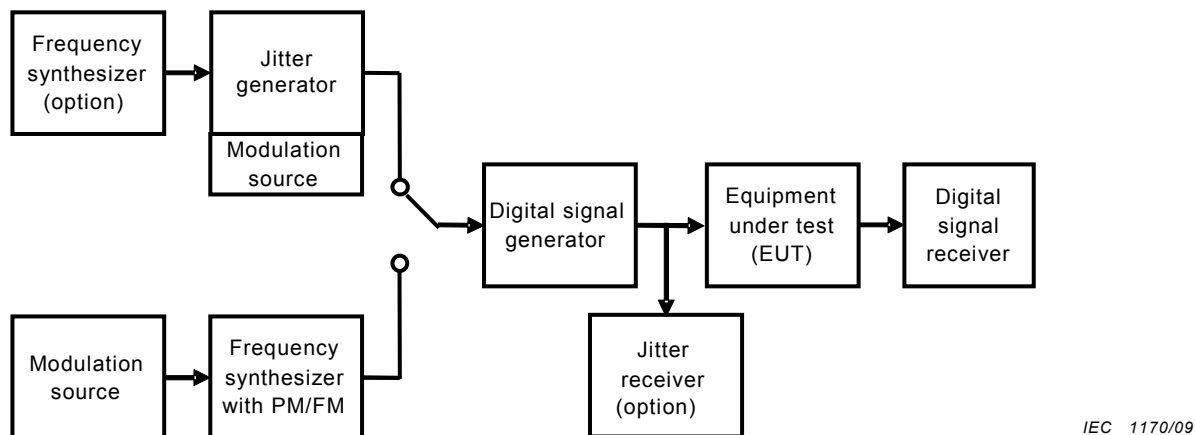


Figure 7 – Jitter tolerance measurement configuration: onset of errors technique

6.4.2 Equipment settings

This technique involves setting a jitter frequency and determining the jitter amplitude of the test signal, which causes the onset of errors criterion to be satisfied. Specifically, this technique requires:

- isolation of the jitter amplitude "transition region" (in which error-free operation ceases);
- one errored second measurement, 30 s in duration, for each incrementally increased jitter amplitude from the beginning of this region; and
- determination of the largest jitter amplitude for which the cumulative errored second count is no more than 2 errored seconds.

The process is repeated for a sufficient number of frequencies so that the measurement accurately represents the continuous sinusoidal input jitter tolerance of the EUT over the applicable jitter frequency range. The test equipment shall be able to produce a controlled jittered signal and measure the resulting errored seconds caused by the jitter on the incoming signal.

6.4.3 Measurement procedure

- Connect the equipment as shown in Figure 7. Verify proper continuity and error-free operation.
- Set the input jitter frequency as desired, and initialize the jitter amplitude to 0 UI peak-to-peak.
- Increase the jitter amplitude in gross increments to determine the amplitude region where error-free operation ceases. Reduce the jitter amplitude to its level at the beginning of this region.
- Record the number of errored seconds that occur over a 30-second measurement interval. Note that the initial measurement shall be 0 errored seconds.
- Increase the jitter amplitude in fine increments, repeating step d) for each increment, until the onset of errors criterion is satisfied.
- Record the indicated amplitude and frequency of the applied input jitter, and repeat steps b) to d) for a sufficient number of frequencies to characterize the jitter tolerance curve.

6.5 Jitter tolerance stressed eye receiver test

6.5.1 Purpose

This test is intended to determine the ability of a receiver or system to operate in the presence of non-ideal signals that are likely to exist in a real communications system. It is similar to a jitter tolerance test. However, unlike a jitter tolerance, the intentional degradation of the signal is not limited to sinusoidal jitter and typically consists of several signal impairment mechanisms.

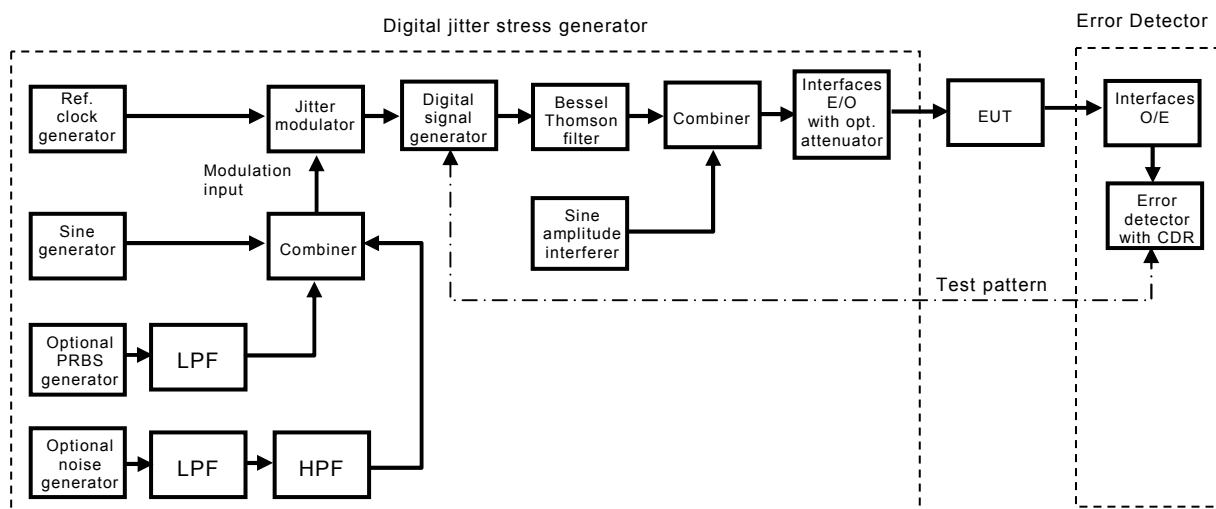
6.5.2 Apparatus

The following apparatus is required:

- digital pattern generator
- digital error detector
- jitter generator
- reference clock

Instrumentation to generate stress may include:

- a sine wave generator (sinusoidal jitter)
- an arbitrary waveform generator (ARB, for periodic jitter)
- a noise source (random jitter) and
- methods to apply the stress signals to the digital data stream (phase/frequency modulatable clock source, delay line modulator, etc.)



IEC 1171/09

Figure 8 – Equipment configuration for stressed eye tolerance test

6.5.3 Sinusoidal jitter template technique

As there may be multiple elements of stress, it is common to vary one stress element while keeping other stress elements fixed. The most common technique is to measure BER over a range of discrete sinusoidal jitter frequencies with other stress elements kept constant.

6.5.3.1 Equipment connection

Figure 8 illustrates the test configuration using a multiple element stressed eye generator. The stress signal is presented to the EUT. The output of the EUT is monitored by an error detector to facilitate the measurement of BER.

6.5.3.2 Equipment settings

This technique is intended to confirm that the EUT is capable of achieving a minimum BER level while receiving a signal with specified levels of impairment. Signal power into the receiver is set to represent the worst case expected level that may occur in normal system operation. Signal impairment (sinusoidal jitter, random jitter, inter-symbol interference, etc.) is also typically set to levels that typically represent the worst case expected levels that may occur in normal system operation.

Signal impairment typically includes some form of sinusoidal jitter. The frequency of the jitter is typically varied over a range of frequencies to verify the receivers' ability to track and follow low frequency jitter. In real systems, low frequency jitter is often larger in magnitude than high frequency jitter. Thus the test process will verify BER at several discrete sinusoidal jitter frequencies, with large jitter magnitudes at low frequencies, decreasing to lower magnitudes at high jitter frequencies.

At each jitter frequency, a BER measurement is made. The process is repeated for a sufficient number of frequencies so that the measurement accurately represents the tolerance of the EUT to continuous sinusoidal input jitter over the applicable jitter frequency range. Other elements of signal stress are kept fixed.

6.5.3.3 Measurement procedure

- a) Connect the equipment to the data output as shown in Figure 8. Verify proper continuity and error-free operation with signal stress elements disabled.
- b) Set the input jitter frequency and magnitude as desired. Enable all other stress elements as required.
- c) Measure the BER and verify it meets acceptable levels.
- d) Increment the jitter frequency and set the appropriate jitter amplitude. Repeat step c).
- e) Repeat step d) for the entire range of sinusoidal jitter frequencies required to verify stressed eye receiver tolerance.

7 Measurement of jitter transfer function

7.1 General

The purpose of this test procedure is to determine the amplitude and frequency relationship between the jitter at the input of an equipment or part of a network and the corresponding output.

7.2 Apparatus

The following apparatus is required:

- jitter generator
- digital signal generator
- spectrum analyzer
- jitter receiver

The frequency synthesizer is optional.

In addition, the enhanced technique requires the following:

- sine-wave generator
- frequency synthesizer with phase modulation
- buffer amplifier

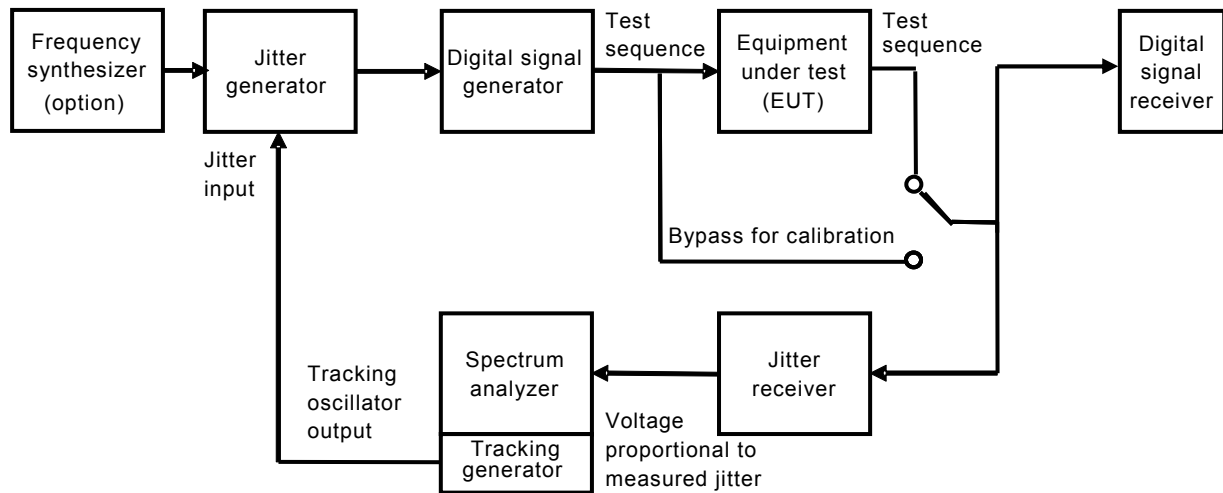
- variable delay line
- dividers
- mixer
- low-pass filter
- phase meter
- voltmeter.

The demultiplexer technique also requires a multiplexer.

7.3 Basic technique

7.3.1 Equipment connection

Connect the equipment as shown in Figure 9, bypassing the EUT. Verify proper continuity, linearity, and error-free operation.



IEC 1172/09

Figure 9 – Measurement of jitter transfer function: basic technique

7.3.2 Equipment settings

Set the frequency range on the spectrum analyzer as desired. Adjust the tracking oscillator output level on the spectrum analyzer to produce a tolerable jitter amplitude over the selected frequency range, which should be large enough to ensure adequate measurement accuracy, yet sufficiently small to preserve linear operation.

Setting the spectrum analyzer bandwidth as narrow as feasible, sweep the desired frequency range and record the 0 dB amplitude reference trace of the test equipment. (Setting a narrow spectrum analyzer bandwidth may allow a reduction in applied jitter amplitude with no loss in measurement accuracy.)

7.3.3 Measurement procedure

Reconnect the EUT as shown in Figure 9. Verify proper continuity, linearity, and error-free operation.

Use the spectrum analyzer to sweep the selected frequency range and record the magnitude of the overall (test equipment and EUT) jitter transfer function. To obtain the EUT jitter transfer function, subtract the 0 dB amplitude reference trace from the overall jitter transfer function.

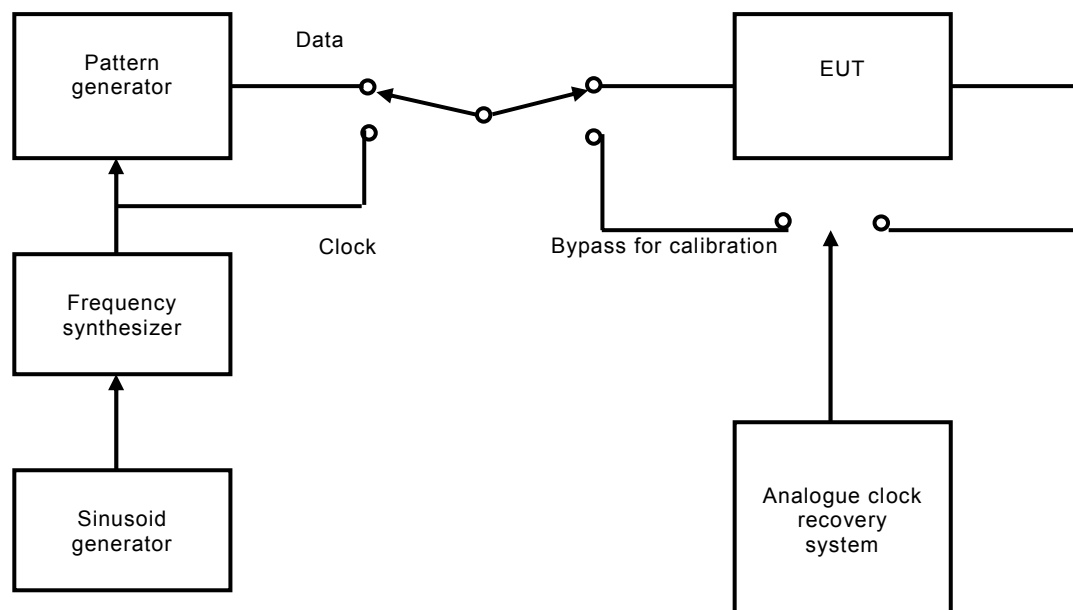
Repeat the measurements for a sufficient number of frequency ranges to characterize the overall frequency range of interest.

7.4 Analogue phase detector technique

An instrumentation grade clock recovery system can be used to determine the magnitude of sinusoidal jitter. The output of the clock recovery internal phase detector can be calibrated to provide an accurate gauge of the jitter going into the EUT and the jitter coming out of the EUT. By stepping the frequency of the jitter over a range of values and taking the ratio of output jitter to input jitter, the jitter transfer result can be obtained. The technique is similar to that of 7.3.1, where the jitter is extracted from the input signal to the EUT (in the calibration step) or output of the EUT. The analogue phase detector acts as the jitter receiver. However, rather than measure the extracted jitter with a spectrum analyzer, instrumentation within the phase detector system, typically some form of calibrated analog to digital converter, measures the jitter magnitude directly.

7.4.1 Equipment connections

Equipment connections vary depending on the class of EUT. A clock recovery system would have data as the input and clock as the output. A repeater would have data in and data out. A clock multiplier would have clock in and clock out. A digital transmitter may use a reference clock input that is multiplied within the EUT to produce a high-speed data output. See Figure 10. A calibration measurement to account for source and receiver unflatness is made by connecting the clock or data line directly to the analogue clock recovery system (see 7.4.2 and 7.4.3). The clock or data line is then connected to the EUT and the EUT output to the clock recovery system.



IEC 1173/09

Figure 10 – Measurement of jitter transfer: analogue phase detector technique

7.4.2 Equipment settings

The sinusoid source is set to an amplitude necessary to produce the desired level of jitter (phase) modulation of the frequency synthesizer. The optimum level is typically EUT dependent. The jitter should not be too large so that either the EUT or clock recovery system is operated in a non-linear range or is forced into an unlocked condition. The frequency of the sinusoid is initially set to a value that is well below the loop bandwidth of the EUT.

7.4.3 Measurement procedure

The frequency of the sinusoid is incremented over a range well below and well above the expected loop bandwidth of the EUT (perhaps 1 decade below to 1 decade above). At each jitter frequency setting, the jitter magnitude is measured and recorded (for both the calibration and EUT measurement). The jitter is measured and recorded for both the calibration and EUT output observations over identical jitter frequency ranges.

7.4.4 Measurement calculations

Jitter transfer is obtained by taking the ratio of the jitter measured at the EUT output to the jitter measured during calibration. A jitter transfer 'curve' is obtained by plotting this ratio at each jitter frequency tested.

8 Measurement of output jitter

8.1 General

The purpose of this test procedure is to measure output jitter at hierarchical interfaces, and intrinsic jitter generated by individual digital equipment. Measurements of output jitter may be in terms of r.m.s. or peak-to-peak amplitudes over designated frequency ranges and may require statistical characterization.

8.2 Equipment connection

The following test equipment is required to perform this test:

- digital signal receiver
- digital signal generator
- jitter receiver
- jitter measurement filter
- voltmeter

The following additional test equipment is optional and intended for use when greater accuracy is required:

- frequency synthesizer
- spectrum analyzer

NOTE This subclause will be expanded in a later issue to call out required attributes and tolerances of test sets.

8.2.1 Equipment settings

The equipment under test is set to generate a repeating data pattern that is the same as the digital signal generator and receiver. An amplitude at which jitter is measured is defined and set.

8.2.2 Measurement procedure

The test sample shall be an item of the fibre optic transmission system, as used under normal operating conditions and having inputs and outputs normally seen by the user of the system. The sample is identified in Figure 11 as equipment under test (EUT).

8.2.2.1 Live traffic

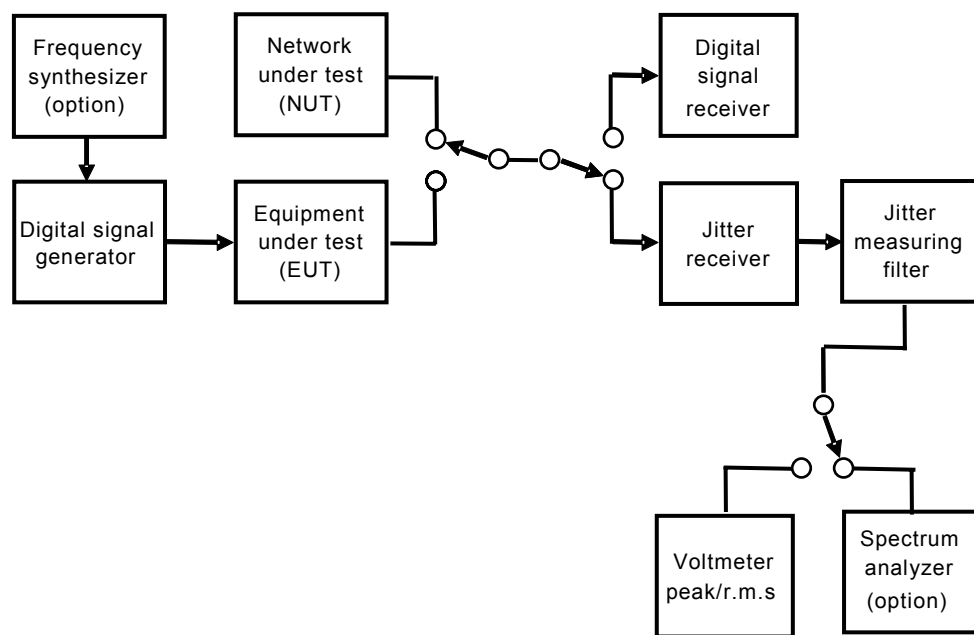
Connect the equipment as shown in Figure 11. Verify proper continuity and error-free operation. Select the desired jitter measurement filter and measure the filtered output jitter, recording the true peak-peak jitter amplitude that occurs during the specified measurement time interval. Repeat procedure for all desired jitter measurement filters.

8.2.3 Controlled data

Connect the equipment as shown in Figure 11, using the digital signal generator to provide a jitter free controlled data pattern (e.g., a pseudorandom bit sequence of length $2^{23}-1$) to the equipment under test. Verify proper continuity and error-free operation.

Select the desired jitter measurement filter and measure the filtered output jitter, recording the true peak-to-peak jitter amplitude that occurs during the specified measurement time interval.

Repeat this procedure for all desired jitter measurement filters.



IEC 1174/09

Figure 11 – Output jitter measurement

9 Measurement of systematic jitter

9.1 Apparatus

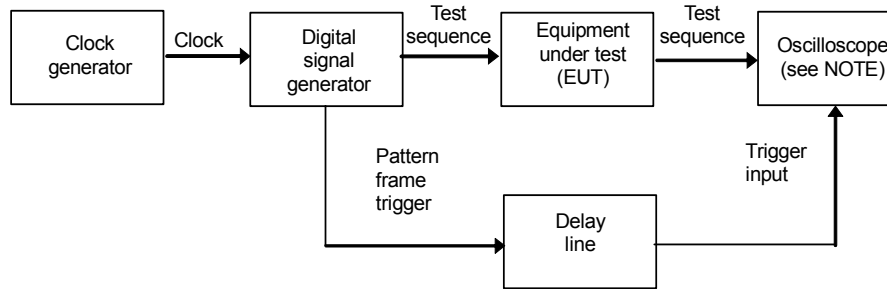
The following apparatus are required:

- clock generator
- digital signal generator
- delay line
- oscilloscope

9.2 Basic technique

9.2.1 Equipment connection

Figure 12 illustrates the test configuration for the systematic jitter measurement basic technique.



IEC 1175/09

NOTE It can replace other phase detector equipment which is based on waveform.

Figure 12 – Systematic jitter measurement configuration: basic technique

9.2.2 Equipment settings

Set the acquisition to average over at least 64 traces on the oscilloscope to eliminate random phase noise. Adjust the phase of the clock rising edge to coincide exactly with digital data signal edges. (This will minimize the effect of non-linearity in the oscilloscope time base.) Measure the time (in UI) between the rising edge of the clock and the corresponding digital signal edge (within $\pm 0,5$ UI).

9.2.3 Measurement procedure

- Set the signal generator to produce a framed test signal as desired pattern.
- Use an oscilloscope, with measurement bandwidth equivalent to a fourth-order Bessel-Thomson filter with a 3 dB attenuation frequency at 0,75 times the bit rate, to extract the data and clock waveforms.
- Set the acquisition to average over at least 64 traces to eliminate random phase noise. Adjust the phase of the clock rising edge to coincide exactly with the digital signal edges. (This will minimize the effect of non-linearity in the oscilloscope time base.) Measure the time (in UI) between the rising edge of the clock and the corresponding digital signal edge (within $\pm 0,5$ UI). See Figure 13. This forms a sequence of pattern-dependent phase values ξ_i .

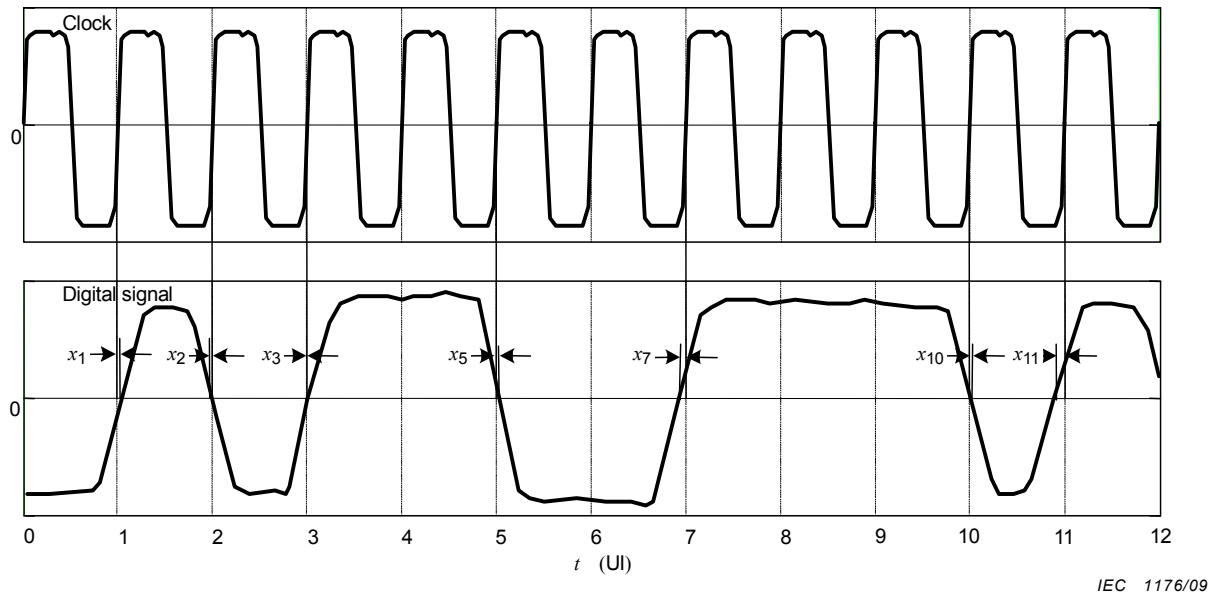


Figure 13 – Measurement of the pattern-dependent phase sequence x_i

- d) If there is no corresponding digital signal edge for a particular i , assign to $x_i = 0$. Measure x_i to cover one period of the digital frame. Then generate mathematically a new sequence by using the following formulae:

$$x'_i = \frac{\sum_{n=1}^{24} x_{i-n}}{\sum_{n=1}^{24} p_{i-n}} \quad (\text{for short test sequences up to PRBS 223-1}) \quad (6)$$

$$x'_i = \frac{\sum_{n=1}^{64} h_n x_{i-n}}{\sum_{n=1}^{64} h_n p_{i-n}} \quad (\text{for long test sequence, PRBS231-1}) \quad (7)$$

where p_i represents the pattern density information.

Assign $p_i = 1$ when a digital edge exists, assign $p_i = 0$ when no transition data is present.

Substitute x'_i values into series x_i where no measured digital signal edge value exists. Substitute x'_i values into series x_i where there is no measured edge value. The h_n series ($n = 1, 2, \dots, 64$) is the coefficient of an LPF with a cut-off frequency of $0,032f_0$.

- e) Filter the sequence x_i mathematically with the appropriate high-pass and low-pass filters to form the sequence of pattern-dependent jitter values y_i .

10 BERT scan technique

10.1 General

Jitter is considered a common root cause to degraded BER. In that system BER performance is often at levels of less than 1 error per trillion bits received ($1E-12$), it can be important to

characterize jitter to similar probabilities. This allows an accurate estimation of the impact of jitter on overall BER performance. One way to visualize the impact of jitter is through the eye diagram. The eye diagram is a display of all the bits of a digital stream overlaid on a common time axis. If the signal is free of jitter, all the rising edges (0 to 1 transitions) will be at their ideal locations in time which implies that they lie directly on top of each other. A similar scenario will exist for falling edges (1 to 0 transitions). When a data stream has jitter, the data edges of the eye diagram will not be at the same time location. The eye diagram will begin to close in the time axis due to both early and late edges.

Typically, the edges that have the largest deviation from ideal also occur at the lowest probabilities. This is especially true when there is a significant random jitter component. If the signal to be tested is input to an error detector of a bit-error-ratio-tester (BERT), and the error detector decision is set to take place in the centre of the eye, no errors are likely to be detected unless the jitter is extremely large. If the decision point is moved away from the centre of the eye, eventually a point will be reached where decisions are made when the bits with the largest jitter are still in transition from one logic state to another. An error will be detected. These errors will generally occur at a very low frequency, as the largest jitter occurs with low probability. As the decision point is continually moved away from centre, the likelihood of errors will increase, as more and more often the decision will be made on or before a data transition. If the BER is recorded as a function of decision circuit time position, varied over a unit interval or bit period, the cumulative distribution function of the jitter can be mapped. This is commonly referred to as a bathtub curve (see Figure 14). In the centre of the unit interval, the probability of errors (which note the probability that a jittered edge has taken place) is extremely low and perhaps immeasurable. Off centre, the probability increases. Thus the function has a shape similar to a tub.

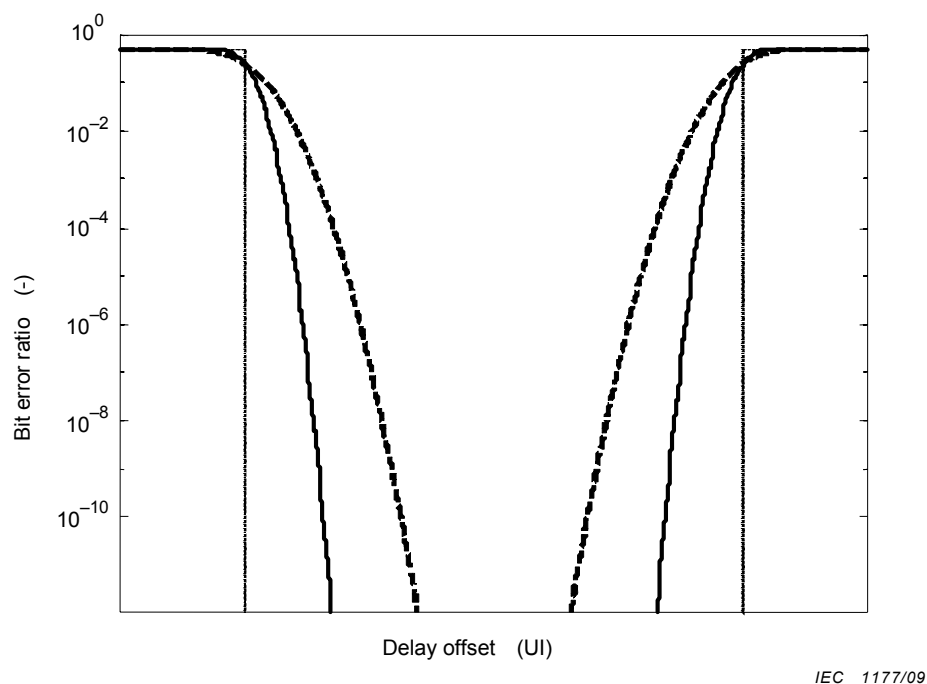


Figure 14 – BERT scan bathtub curves (solid line for low jitter, dashed line for high jitter)

This technique allows the direct measurement of jitter probabilities to extremely low levels, such as edge positions that occur at a one in a trillion (or lower) rate. However, when trying to assess jitter to these low probabilities, many trillions of bits shall be observed. The measurement can take many hours to perform, as the error rate for any time position cannot be accurately determined until a significant number of errors have been measured. Measurement times in the regions of the eye diagram are thus long, while measurement times in the centre regions of the eye are generally low.

10.2 Apparatus

The following apparatus is required:

- error detector
- clock signal representing ideal jitter performance

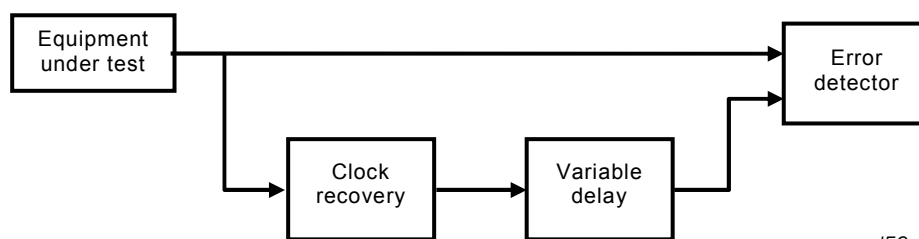
10.3 Basic technique

10.3.1 Equipment connection

The signal to be analyzed is input to the data port of the error detector. The ideal clock signal is input to the clock port of the error detector.

10.3.2 Equipment settings

The error detector is configured to determine the ideal sampling point within the data eye diagram in both amplitude and time. The clock signal provides the reference that determines when the error detector decision takes place. All jitter is measured relative to this reference. If the clock is approximately jitter free and synchronous to the nominal rate of the data stream, the measured jitter will include all timing deviations of the signal. In contrast, if a clock signal is derived from the data stream itself through some clock extraction system, jitter that is common to both the clock and the data will not be observed in the BERT scan (Figure 15). The observed jitter range will be the complement of the jitter bandwidth of the clock extraction system. In that jitter bandwidth is generally a low-pass function, using an extracted clock for a timing reference generally produces a high-pass result. Jitter above the jitter bandwidth is observed. This is sometimes desired, as low frequency jitter is often of minor concern and easily accommodated by a system receiver.



IEC 1178/09

Figure 15 – Equipment setup for the BERT scan

10.3.3 Measurement process

With the error detector sampling point at the ideal amplitude/time position, and aligned to the data pattern, the sampling point is adjusted to one half unit interval earlier in time. A significant number of errors is detected and the BER is determined and recorded for this time relative time value. The sample point time is increased a small fraction of a unit interval towards the centre of the eye and the next BER-time position data pair is recorded. With each successive measurement, the BER typically is reduced. The process is repeated until the BER becomes too low to be measured or the necessary BER threshold is obtained (e.g. 1E-12 or lower). The sampling point is then set to one half unit interval later than the ideal sampling time. The BER is determined and the above process is repeated, except that the sampling point is reduced towards the centre of the eye.

For decreased test times, low BER values can be extrapolated from measured values at higher BER.

11 Jitter separation technique

11.1 General

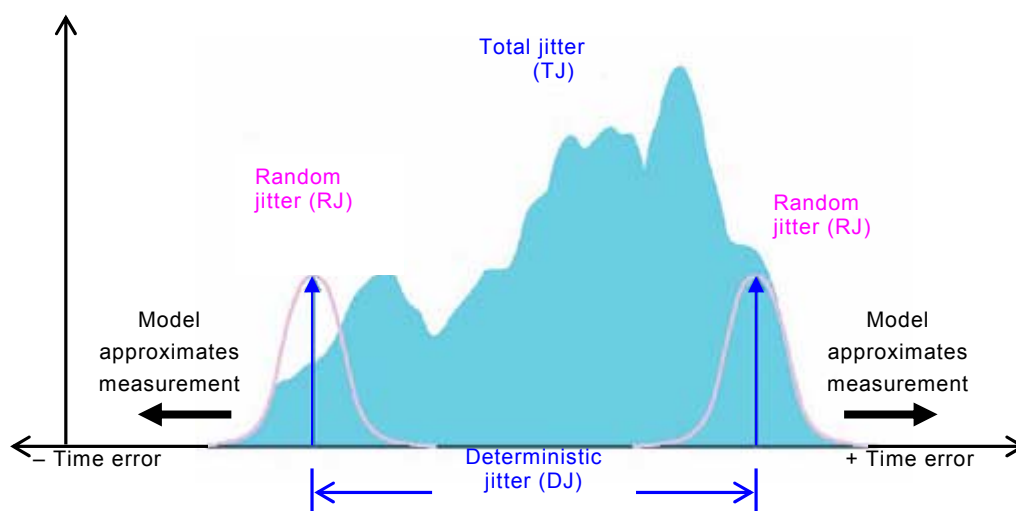
The direct assessment of signal performance to capture events with extremely low probabilities implies a very large measurement population which in turn implies a lengthy measurement time (see BERT scan technique). Time efficient estimation techniques based upon reduced data sets are susceptible to large measurement errors since jitter originates from a variety of sources with a variety of probability distributions. One technique that can provide accurate estimations of total jitter to include events with very low probabilities, without the lengthy time required by a direct measurement, is the jitter separation technique. The basis of this technique is to directly measure the high probability components of the jitter and accurately estimate the distribution of low probability jitter components. A model is then constructed using the various components to estimate the distribution of the aggregate and predict the extent of the jitter to any probability desired.

A practical approach to the jitter separation technique is to use a test signal with a repeating test pattern. The average position of each edge in the pattern is determined and compared to its expected position. This yields the data dependent or correlated jitter. Observing any specific edge in the pattern and determining the distribution of edge locations yields the uncorrelated jitter. This would include any random jitter and periodic jitter that is independent of position in the data pattern. In other words, this jitter appears similarly on any edge when that edge is observed multiple times as the pattern is repeated. The periodic uncorrelated jitter will have a bounded distribution. That is, its magnitude is limited. Truly random jitter is unbounded. It generally has a Gaussian distribution that extends infinitely within the physical limits of the system. Thus it can be accurately described through its standard deviation.

To determine the standard deviation of the random jitter, it shall be separated from the uncorrelated periodic jitter. Various techniques have been implemented including viewing the uncorrelated jitter population in the frequency domain and removing periodic spectra through post-processing, or by curve fitting the uncorrelated jitter population.

If the aggregate jitter for a device were completely random, estimating the total jitter to include events with low probabilities could be achieved by characterizing the jitter population and determining the standard deviation. The standard deviation would then be multiplied appropriately to determine jitter to a specific probability level. For example, in order to determine the magnitude of the jitter to a probability of 10^{-12} , the standard deviation is multiplied by approximately 14. This indicates the critical need to accurately determine the standard deviation of the random jitter. Any error can significantly alter the estimate of the total jitter due to the typically large multipliers involved.

Jitter is rarely composed of random components alone. Estimating total jitter requires an analysis that includes all the jitter components. A common approach is the dual-Dirac model (Figure 16). In this approach, the standard deviation of the random jitter is determined. The random jitter distribution is then constructed. This distribution is then split apart and positioned according to the effective magnitude of the aggregate deterministic jitter components. It should be noted that the effective deterministic jitter may not be identical to the actual magnitude of the aggregate deterministic jitter, but instead is a derived value that leads to the best estimate of total jitter when combined in the model with the random jitter.



IEC 1179/09

Figure 16 – Dual-Dirac jitter model

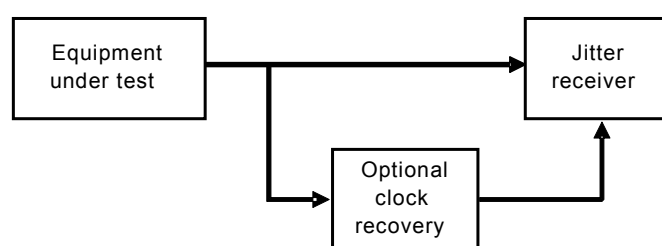
With a model, the jitter can be estimated to extremely low probabilities, without having to perform measurements in low population regions of the signal. Keys to accurate measurements include not confusing low probability periodic jitter as random jitter, low residual jitter and noise in the test equipment, and an efficient method to obtain sufficient populations to create an accurate model.

11.2 Apparatus

The following equipment is required for the measurement:

- a device that can locate the time location of the edges of a digital bit stream (oscilloscope, time interval analyzer, BERT)
- a clock representing the ideal position of data edges.

The basic equipment setup is shown in Figure 17.



IEC 1180/09

Figure 17 – Equipment setup for jitter separation measurement

11.3 Equipment connections

The signal under test is connected to the edge measuring instrument. A clock signal is also connected as a timing reference. In some cases, this clock is derived from the signal under test.

11.4 Equipment settings

Typically the device under test is set to generate a repeating data pattern. An amplitude threshold at which time is measured is defined and set. Typically this is the middle level of the signal amplitude. If a clock signal timing reference is to be derived from the signal under test,

the jitter bandwidth of the clock extraction system should be set to provide the correct filtering of the observed jitter (see 9.2.2).

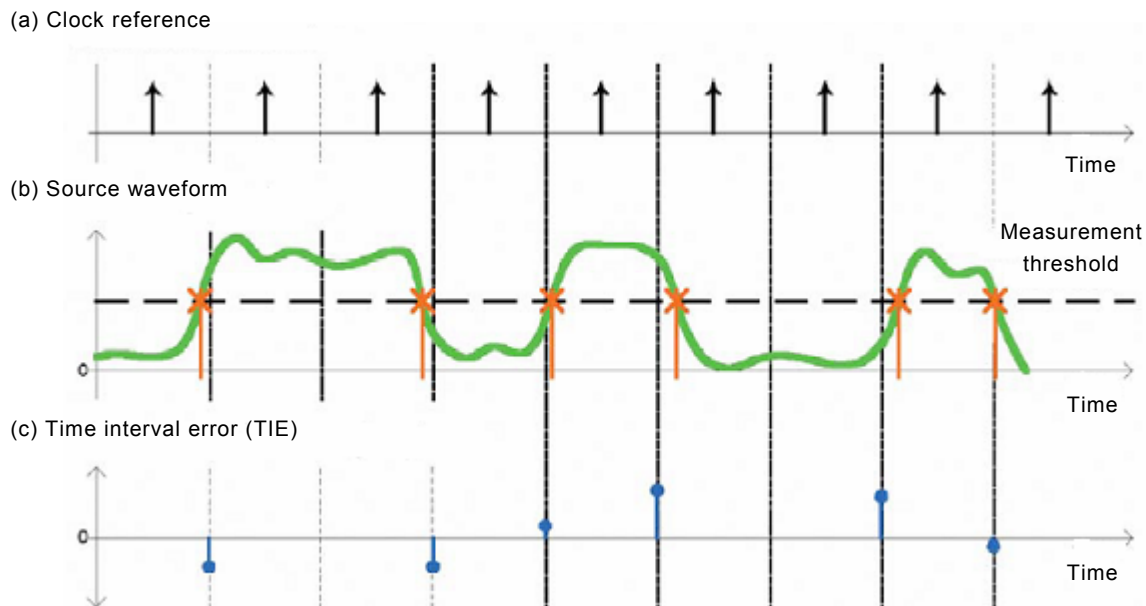
11.5 Measurement procedure

11.5.1 Sampling oscilloscope

All edges in the data pattern are located and recorded for average position versus ideal. The time separation between the overall earliest edge position (versus ideal) to the latest edge position (versus ideal) represents the data dependent jitter. Uncorrelated jitter is obtained from examining any edge in the pattern and obtaining a population of edge positions versus ideal. Samples are taken periodically to allow transformation into the frequency domain. In the frequency domain, the noise floor of the spectrum represents the random jitter. Discrete spectral lines represent any periodic components. Integration of the noise floor with the line spectra removed yields the random jitter, with further processing yielding the standard deviation. The energy of the line spectra yields the periodic jitter. Convolution of the various components yields an overall population. The dual-Dirac model is then adjusted to optimally fit the reconstructed jitter population. The model can then provide total jitter magnitude to the desired probability. (Details of the signal processing exceed the scope of this standard.)

11.5.2 Real-time oscilloscope

The real-time oscilloscope is capable of capturing an entire data pattern in a single waveform acquisition. Jitter analysis is based on variations in time-displacement of voltage transitions of a serial data waveform relative to a specified time reference. Effectively, the measurement process analyzes time interval error (TIE) (Figure 18). TIE is a discrete-time function of time error versus time. The time reference (clock reference) used for TIE measurements can be defined in many different ways. One time reference commonly used for TIE measurements is a constant-frequency square wave with frequency and phase that has been best-fit to the acquired waveform record. Sometimes the voltage transitions of a second source (such as a reference clock signal) are used as the time reference. (See 9.2.2).



IEC 1181/09

Figure 18 – Measurement of time interval error

In order to isolate the jitter that is correlated to the data pattern, the analysis shall first calculate the TIE function of the jitter and associate each TIE value with a specific bit in the source waveform's logical bit sequence. This is done by extracting the logical bit sequence from the source waveform and determining the length in bits of its periodic pattern. Next, the

original TIE function is decimated into sub-sampled TIE functions, where all of the values in each sub-sampled function correspond to a specific bit within the pattern.

Each of these sub-sampled TIE functions is then transformed into the frequency domain using a fast Fourier transform (FFT). DDJ is now separated from the rest of the jitter because the first value of each jitter spectrum (DC component) is equal to the DDJ for that particular bit of the repeating bit pattern.

Once the DDJ has been subtracted from all of the TJ spectrums, the remaining jitter spectrums are entirely comprised of RJ and PJ. The first step in separating PJ from RJ is to calculate the PSD (power spectral density) of all the remaining RJ/PJ spectrums. All of the individual RJ/PJ spectrums are averaged together (as well as averaged with spectrums from previous acquisitions) to form an APSD (averaged PSD). At this point, all of the APSD's frequency components that have significantly large magnitudes are removed, because they could potentially contain PJ. The remaining frequency components of the APSD are then combined to obtain the root-mean-square (rms.) value of RJ.

DJ(d-d) and PJ(d-d) are both determined by fitting the dual-Dirac model described above to measured histograms. The dual-Dirac model can be fitted to a PDF using various methods, many of which solve for the Gaussian component and the bimodal component simultaneously. Once the model is constructed, total jitter values to the desired probability of an event can be derived.

12 Measurement of wander

12.1 Apparatus

The following apparatus is required:

- reference clock generator
- digital signal generator
- wander receiver

12.2 Basic technique

12.2.1 Equipment connection

Due to the low frequency of the phase variations to be evaluated (refer to definition 3.21), wander is a quantity, which requires a special test configuration. When performing jitter measurements, the required reference-timing signal is normally produced locally—by means of a phase-locked loop (PLL) within the test set; it is derived from the average phase of the signal to be measured. Such a PLL cannot be realized to cope with the requirements of wander measurements.

Therefore, wander measurements always require an external reference clock signal of adequate stability.

12.2.1.1 Synchronized wander measurements

Figure 19 illustrates the test configuration for the synchronized wander measurement basic technique.

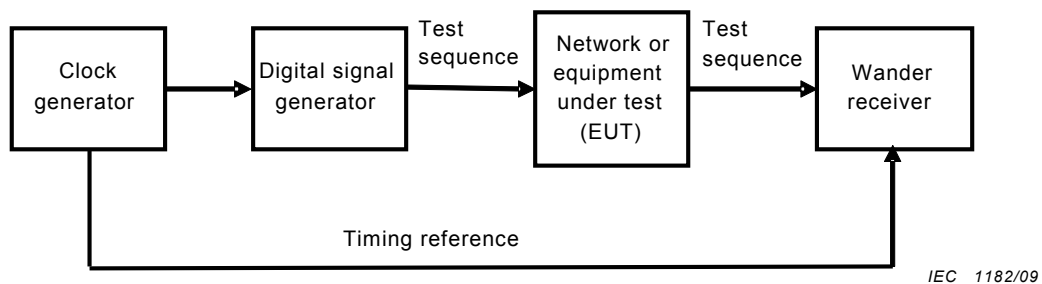


Figure 19 – Synchronized wander measurement configuration

This configuration is applicable if the timing signals required to perform the measurement can be derived from a common reference clock. This means that only loop measurements, where input and output ports of the unit-under-test are accessible at the same location, can be carried out in this way. In this set-up, the measurement result is not affected by phase variations of the reference clock. Thus, the requirements on the stability of the reference clock are not very high and are achievable in portable test instrumentation.

12.2.1.2 Non-synchronized wander measurements

Figure 20 illustrates the test configuration for the non-synchronized wander measurement basic technique. This configuration is applicable to wander measurements in cases where both input and output ports of the network or equipment under test are not available at the same location (e.g. end-to-end measurements). In this set-up, any frequency/phase drift of the two clocks involved in the measurement affects the measurement result. This means that the stability of the two clocks has to be at least one order of magnitude better than the quantity to be measured. Such reference clocks may not be provided in portable test instrumentation in which case synchronization to an external reference is required.

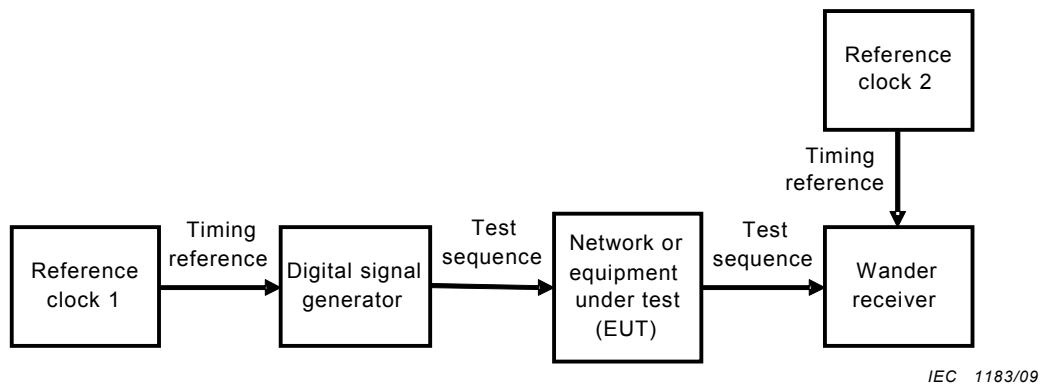


Figure 20 – Non-synchronized wander measurement configuration

12.2.2 Equipment settings

In order to calculate and estimate the various wander parameters defined in Clause 3, TIE is treated as a sampled parameter since continuous knowledge of the time interval error is not practically attainable.

The maximum sampling time τ_0 , of TIE shall be: 1/30 s.

Wander shall be measured through an equivalent 10 Hz, first-order, low-pass measurement filter and with the following characteristics: The low-pass measurement filter has a single-order characteristic and a roll-off of –20 dB/decade. The –3 dB point of the measurement filter shall be at a frequency of 10 Hz.

12.2.3 Measurement procedure

The test sample shall be an item of the fibre optic transmission system, as used under normal operating conditions and having inputs and outputs normally seen by the user of the system. The sample is identified in Figure 19 and 20 as network or equipment under test (EUT).

13 Measurement of wander TDEV tolerance

13.1 Intent

The intent of this test procedure is to measure wander TDEV tolerance in terms of the wander TDEV amplitude that, when applied to an equipment input, causes a designated degradation of error performance. Wander TDEV tolerance is a function of the amplitude and integration time of the applied wander TDEV mask.

13.2 Apparatus

The following apparatus is required:

- external reference clock source
- wander generator (TDEV)
- clock generator

The wander receiver (TDEV) is optional.

13.3 Basic technique

The onset of errors criterion for wander TDEV tolerance measurements is defined as the largest amplitude of wander TDEV mask that causes a cumulative total of more than 2 errored seconds, where these errored seconds have been summed over 12 times of TDEV integration time measurement intervals of increasing wander amplitude.

13.4 Equipment connection

Subclauses 13.4.1 and 13.4.2 contain information on test configurations for wander TDEV tolerance measurements that are in accordance with ITU-T Recommendation G.813.

13.4.1 Wander TDEV tolerance measurement for the test signal of EUT

Figure 21 illustrates the test configuration for the test signal of EUT basic technique. The optional wander receiver (TDEV) is used to verify the amplitude of generated wander TDEV.

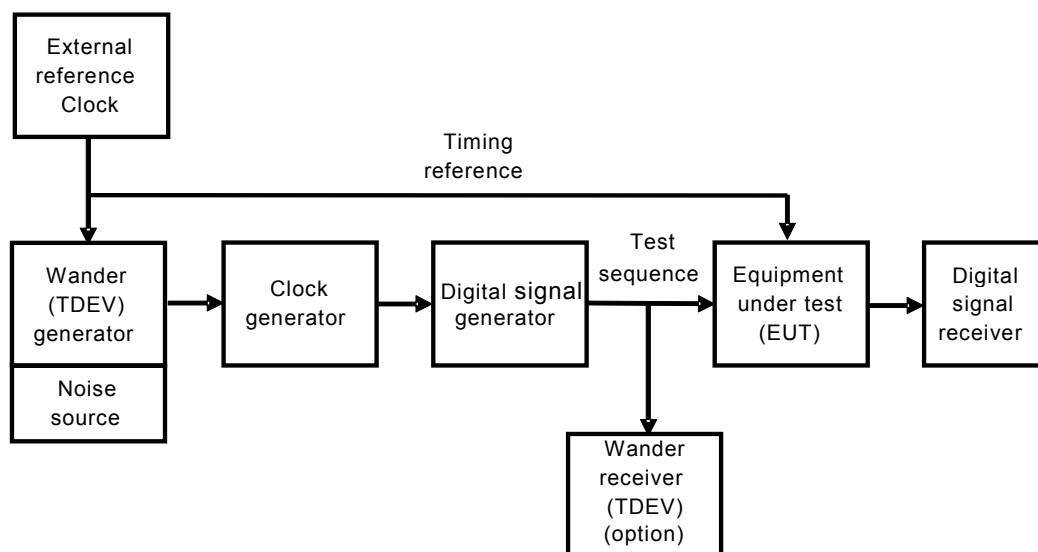
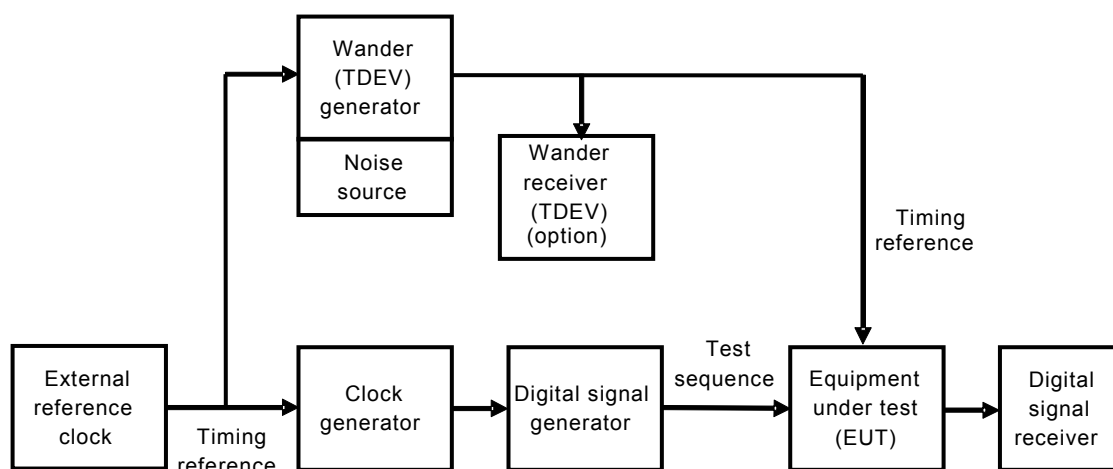


Figure 21 – Wander TDEV tolerance measurement configuration for the test signal of EUT

13.4.2 Wander TDEV tolerance measurement for timing reference signal of EUT

Figure 22 illustrates the test configuration for the timing reference signal of the EUT basic technique. The optional wander receiver (TDEV) is used to verify the amplitude of generated wander TDEV.



IEC 1185/09

Figure 22 – Wander TDEV tolerance measurement configuration for the timing signal of EUT

13.5 Equipment settings

This technique involves setting an integration time of TDEV for measurement interval and determining the wander TDEV amplitude of the test signal which causes the onset of errors criterion to be satisfied. Specifically, this technique requires:

- isolation of the wander TDEV amplitude "transition region" (in which error-free operation ceases);
- one errored second measurement, 12 times of TDEV integration time in duration, for each incrementally added wander TDEV amplitude from the beginning of this region;
- determination of the largest wander TDEV mask for which the cumulative errored second count is no more than 2 errored seconds.

13.6 Measurement procedure

- Connect the equipment as shown in Figure 21 or Figure 22. Verify proper continuity and error-free operation.
- Set and initialize the wander TDEV amplitude to 0ns peak-to-peak.
- Set the wander TDEV amplitude as desired TDEV mask, which is specified in applicable standard.
- Record the number of errored seconds that occur over 12 times of integration time of TDEV measurement interval. Note that the initial measurement shall be 0 errored seconds.
- Increase the wander TDEV amplitude in fine increments, repeating step d) for each increment, until the onset of errors criterion is satisfied.
- Record the indicated wander TDEV to characterize the wander TDEV tolerance curve.

14 Measurement of wander TDEV transfer

14.1 Apparatus

The following apparatus are required:

- external reference clock source
- wander generator (TDEV)
- clock generator
- digital signal generator
- wander receiver (TDEV)

14.2 Equipment connection

Subclauses 14.2.1 and 14.2.2 contain information on test configurations for wander and TDEV transfer measurements that are in accordance with ITU-T Recommendation G.813.

14.2.1 Wander TDEV transfer measurement for the test signal of EUT

Figure 23 illustrates the test configuration for the test signal of EUT basic technique.

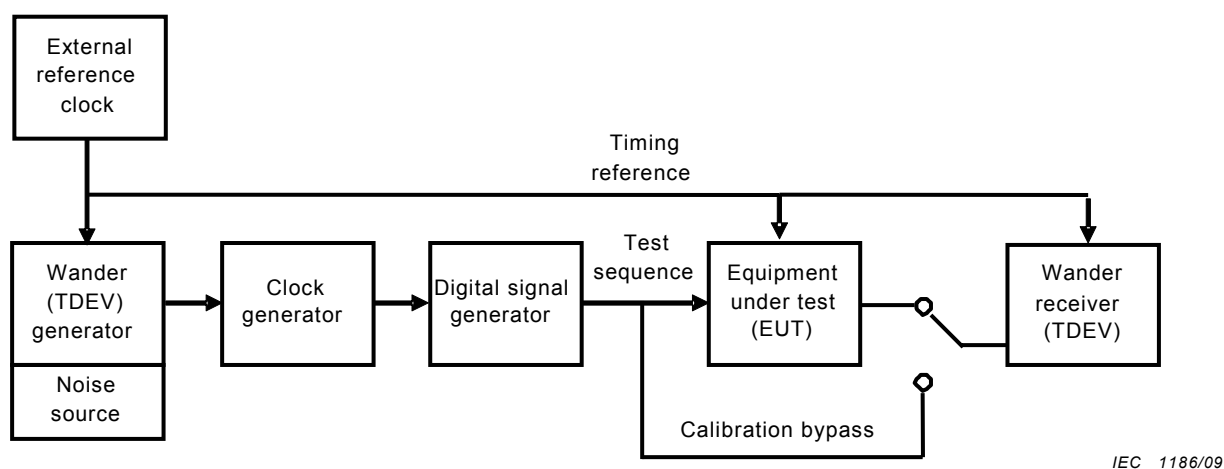


Figure 23 – Wander TDEV transfer measurement configuration for the test signal of EUT

14.2.2 Wander TDEV transfer measurement for timing reference signal of EUT

Figure 24 illustrates the test configuration for the timing reference signal of EUT.

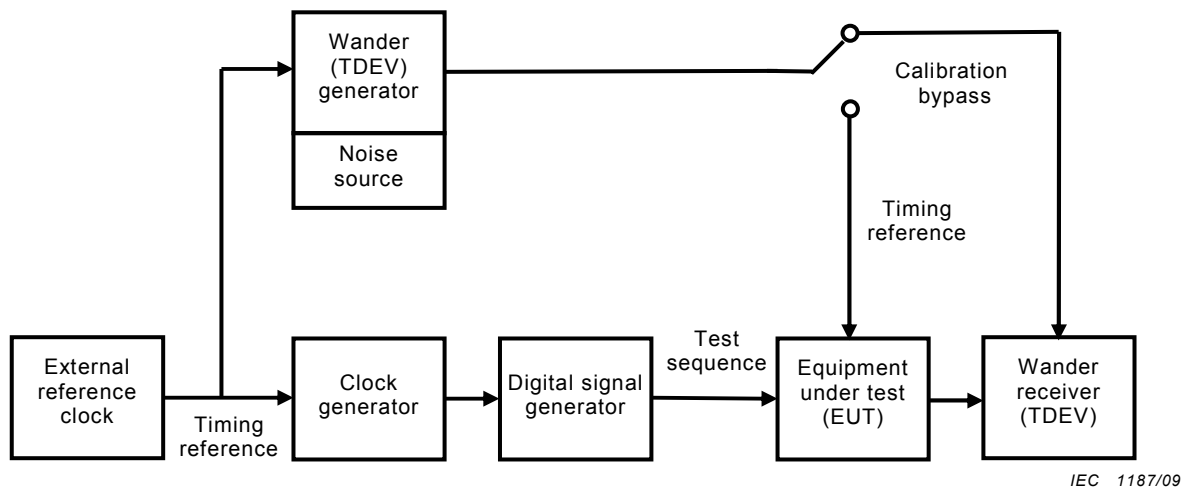


Figure 24 – Wander TDEV transfer measurement configuration for the timing signal of EUT

14.3 Equipment settings

Set tolerable wander amplitude over the specific integration time range, which should be large enough to ensure adequate measurement accuracy, yet sufficiently small to preserve linear operation.

14.4 Measurement procedure

- Perform a wander TDEV tolerance measurement of the EUT over 12 times of integration time of TDEV measurement interval.
- Connect the equipment as shown in Figure 23 or 24, bypassing the EUT. Verify proper continuity, linearity, and error-free operation.
- Set specific wander TDEV mask over the specific integration time range.
- Measure wander TDEV with adding the desired TDEV mask, and record the difference between measured TDEV result and reference TDEV mask which is ideal adding TDEV mask as reference trace of the test equipment.
- Re-connect the EUT as shown in Figure 23 or 24. Verify proper continuity and error-free operation.
- Measure wander TDEV with adding the desired TDEV mask same as step d). Record the wander TDEV results as Wander TDEV transfer function of EUT.
- To obtain the magnitude of the EUT wander TDEV transfer function, subtract the reference trace (step d) from the overall wander TDEV measurement result.

15 Test results

15.1 Mandatory information

The following information shall be provided when the tests have been completed:

- full identification of the equipment under test (EUT)
- full title of test performed
- identification of the test method used
- identification of the test procedure used
- statement of the operating conditions
- statement of environmental conditions
- test results

- date and time when the test was performed.

15.2 Available information

The following information shall also be available:

- full identification of the test equipment used
- identification of electrical patch cords
- identification and performance details of any optical patch cords
- statement of measurement uncertainty
- calibration details of test equipment
- names of test personnel.

Bibliography

- [1] P. Bylinski and D.G.W. Ingram: *Digital transmission Systems 2nd Edition, IEE Telecommunications Series 4, 1980*
- [2] W. R. Bennet: *Bell System Technical Journal 1958, 37*
- [3] C.J. Byrne, B.J. Karafin and D.B. Robinson Jr.: *Systematic jitter in a chain of digital regenerators, Bell System Technical Journal 1963, 42*

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